Letter Vol. 45, No. 21 / 1 November 2020 / Optics Letters 5937

Optics Letters

A flexible numerical calculation method of angular spectrum based on matrix product

WANLI ZHAO,[1] CHENLU WEI,[1] CAOJIN YUAN,[2] CHENLIANG CHANG,[2,3] JUN MA,[1,*] AND RIHONG ZHU[1]

[1]School of Electronic and Optical Engineering, Nanjing University of Science and Technology, No. 200 Xiaolingwei Street, Nanjing, Jiangsu 210094, China
[2]Key Laboratory for Opto-Electronic Technology of Jiangsu Province, Nanjing Normal University, Nanjing 210023, China
[3]Department of Bioengineering, University of California, 410 Westwood Plaza, Los Angeles, California 90095, USA
*Corresponding author: majun@njust.edu.cn

Received 14 August 2020; revised 18 September 2020; accepted 18 September 2020; posted 21 September 2020 (Doc. ID 405257); published 23 October 2020

Fast Fourier transform (FFT) is the most commonly used mathematical method in numerical calculation, and the FFT-based angular spectrum method (ASM) is also used widely in diffraction calculation. However, the frequency and spatial sampling rules in FFT limit the effective propagation distance and the observation window range of ASM. A novel method for calculating the angular spectrum based on the matrix product is proposed in this Letter. This method realizes the fast calculation of discrete Fourier transform (DFT) based on the matrix product, in which the sampling matrix is orthogonally decomposed into two vectors. Instead of FFT, angular spectrum diffraction calculation is carried out based on the matrix product, which is named the matrix product ASM. The method in this Letter uses a simple mathematical transformation to achieve maximum compression of the sampling interval in the frequency domain, which significantly increases the effective propagation distance of the angular spectrum. Additionally, the size of the observation window can be enlarged to obtain a wider calculation range by changing the spatial sampling of the output plane.

https://doi.org/10.1364/OL.405257

Scalar diffraction theory describes the propagation process of coherent light waves quantitatively [1,2]. Numerical calculation of light wave propagation based on scalar diffraction is widely used in many applications, such as digital holography [3], phase recovery [4], compressed sensing [5], and beam quality analysis [6]. The angular spectral integration is a solution that strictly satisfies the Helmholtz equation [1], and the fast Fourier transform (FFT)-based angular spectrum method (ASM) is widely used because of its fast computing speed [2]. The advantage of FFT-based ASM is that it performs well in the near field, and it can be used for the tilted system [7]. However, the FFT-based ASM is not suitable for far field due to the problem of undersampling of frequency transfer function [8], and the size of the observation plane cannot be changed flexibly. In the far field, it is possible that the observation plane cannot obtain the entire light field. These defects limit the effective calculation range of the angular spectrum in three-dimensional (3-D) space.

In recent years, various methods have been proposed to improve the FFT-based ASM. There are two main ideas to solve the problem of insufficient sampling in the frequency domain. One is to compress the sampling interval to satisfy the sampling conditions, and the other is to limit the frequency range and just calculate the part that satisfies the sampling conditions. A typical method corresponding to the former is the wide window ASM (WWASM) [9], which uses the reciprocal relationship between the frequency sampling interval and the size of the input window in FFT, and compresses the frequency sampling interval by expanding the input window size. The disadvantage is that as the distance increases, the amount of calculation and memory (RAM) will rise sharply. The typical method corresponding to the latter is the band-limited angle spectrum method (BLAS) [10], which only retains the part that meets the sampling conditions. The main problem of the BLAS is that the effective bandwidth at a long distance is too narrow. The above two methods are contradictory in calculation accuracy and calculation amount. The non-uniform sampling fast Fourier transform (FFT) ASM [11,12] uses NUFFT instead of FFT to calculate the integral of the angular spectrum. This method significantly widens the effective calculation bandwidth of the angular spectrum and improves the calculation accuracy at long distances. However, the introduction of convolution operation makes angular spectrum calculation more complicated.

The above methods are a kind of "compromise" to the FFT sampling rules, which sacrifices the simplicity of calculation in order to change the sampling in ASM. In this Letter, matrix product operation is used to implement the fast calculation of discrete Fourier transform (DFT), and the sampling in the frequency and space domains can be controlled freely by setting the sampling vector. The calculation method of an angular spectrum based on the matrix product can flexibly change the frequency sampling range to compress the frequency domain sampling interval according to the sampling conditions of transfer function, and maximize the expansion of the effective bandwidth under limited calculation. Based on this, the effective propagation distance of an angular spectrum is greatly extended. In addition, by setting the spatial sampling vector, the size of the observation window can be expanded flexibly to broaden the calculation range of the observation plane.

The ASM begins with a convolution theorem according to

$$U(x', y') = \mathcal{F}^{-1}\left\{\mathcal{F}\{U_0(x, y)\} \cdot H(f_x, f_y)\right\}, \quad (1)$$

where $\mathcal{F}$ and $\mathcal{F}^{-1}$ represent Fourier transform and inverse transform, respectively; U_0 is the complex amplitude of the input plane; x, y are the coordinates of the input plane; "$\cdot$" represents elementwise multiplication; x', y' represent the coordinates of the observation plane; and f_x, f_y are frequency domain coordinates. The transfer function for free-space propagation is given by

$$H(f_x, f_y) = \exp\left[i\frac{2\pi}{\lambda}d\sqrt{1-(\lambda f_x)^2-(\lambda f_y)^2}\right]. \quad (2)$$

As f_x increases, the peak of $|\partial H/\partial f_x|$ is approximately linearly increasing and is proportional to the diffraction distance d. In the FFT, the sampling interval in the frequency domain is $1/L$ so, when $L < |\partial H/(\pi\partial f_x)|$, the sampling conditions in the frequency domain are no longer satisfied, and the output will be aliased. Both the WWAS [9] and the BLAS [10] are designed to solve this problem. The WWAS increases the calculation window L to satisfy the above condition, while the band limit method directly sets the frequency that does not satisfy the sampling condition to 0. The former requires a large amount of calculation and memory (RAM), and the latter has low calculation accuracy at a long distance. One of the main reasons for the low accuracy of the band-limited ASM is that large numbers of frequency sampling points are wasted [12], so the effective bandwidth is limited. If all the sampling points in the frequency domain can be fully utilized, the effective bandwidth can be accordingly broadened without increasing the amount of calculation, and the calculation accuracy will be improved. However, the sampling range and sampling interval in the frequency domain cannot be manipulated in FFT, which means that the corresponding effective bandwidth is definitely limited.

The two-dimensional (2-D) DFT can be represented as a triple-matrix product [13]. The matrix product form can be improved as

$$\boldsymbol{F}(\boldsymbol{f_x}, \boldsymbol{f_y}) = \frac{1}{MNK_{f_x}K_{f_y}S_1S_2}\boldsymbol{M}^T_{\boldsymbol{y},\boldsymbol{f_y}}\boldsymbol{f}(x, y)\,\boldsymbol{M}^T_{\boldsymbol{f_x},\boldsymbol{x}}, \quad (3)$$

where $\boldsymbol{M}_{\boldsymbol{y},\boldsymbol{f_y}} = \exp(-i2\pi\,\boldsymbol{y}^T\boldsymbol{f_y})$, $\boldsymbol{M}_{\boldsymbol{f_x},\boldsymbol{x}} = \exp(-i2\pi\,\boldsymbol{f}^T_{\boldsymbol{x}}\boldsymbol{x})$, $\boldsymbol{f}(\boldsymbol{x}, \boldsymbol{y})$ is the original function in discrete form, $\boldsymbol{F}(\boldsymbol{f_x}, \boldsymbol{f_y})$ is the corresponding spectrum function, and $\boldsymbol{f_x}, \boldsymbol{f_y}$ are as follows:

$$\boldsymbol{f_x} = \left[-\frac{n}{2}, -\frac{n}{2}+1, \ldots, \frac{n}{2}-1\right]\frac{2fx_{\mathrm{MAX\_}fft}}{K_{f_x}n}, \quad (4)$$

$$\boldsymbol{f_y} = \left[-\frac{m}{2}, -\frac{m}{2}+1, \ldots, \frac{m}{2}-1\right]\frac{2fy_{\mathrm{MAX\_}fft}}{K_{f_y}m}. \quad (5)$$

In Eqs. (4) and (5), $fx_{\mathrm{MAX\_}fft} = 1/2\Delta x$, $fy_{\mathrm{MAX\_}fft} = 1/2\Delta y$, $n = S_1N$, $m = S_2M$, N and M are the number of original function samples, and $K_{f_x}, K_{f_y}, S_1, S_2$ are sampling control coefficients which can be altered to change the sampling rules in frequency domain. Similarly, the inverse DFT can be written as

$$\boldsymbol{f}(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{M}_{\boldsymbol{y}',\boldsymbol{f_y}}\boldsymbol{F}(\boldsymbol{f_x}, \boldsymbol{f_y})\,\boldsymbol{M}_{\boldsymbol{u},\boldsymbol{x}'}. \quad (6)$$

In Eq. (6),

$$\boldsymbol{x}' = \left[-\frac{n'}{2}, -\frac{n'}{2}+1, \ldots, \frac{n'}{2}-1\right]\Delta x', \quad n' = r_1N, \quad (7)$$

$$\boldsymbol{y}' = \left[-\frac{m'}{2}, -\frac{m'}{2}+1, \ldots, \frac{m'}{2}-1\right]\Delta y',\ m' = r_2M. \quad (8)$$

Therefore, the ASM can be written as

$$\boldsymbol{U}(\boldsymbol{x}', \boldsymbol{y}') = \frac{1}{MNK_{f_x}K_{f_y}S_1S_2}\boldsymbol{M}_{\boldsymbol{y}',\boldsymbol{f_y}} \times\left\{\left[\boldsymbol{M}_{\boldsymbol{y},\boldsymbol{f_y}}{}^T\boldsymbol{U}_0(\boldsymbol{x}, \boldsymbol{y})\boldsymbol{M}_{\boldsymbol{f_x},\boldsymbol{x}}{}^T\right]\cdot\boldsymbol{H}(\boldsymbol{f_x}, \boldsymbol{f_y})\right\}\boldsymbol{M}_{\boldsymbol{f_x},\boldsymbol{x}'}. \quad (9)$$

In this Letter, this method is called the matrix product ASM (MPASM). When the relationship $S_1 = S_2 = 1$, $K_{f_x} = K_{f_y} = 1$, $r_1 = r_2 = 1$, $\Delta x' = \Delta x$, $\Delta y' = \Delta y$ holds, it is consistent with FFT-based ASM. Compared with the FFT method, the advantage of the MPASM is that the sampling rules in the frequency and space domains can be altered by controlling different coefficients. Based on this, the 3-D expansion of the effective calculation range of the angular spectrum is complemented.

The main sampling issue in ASM focuses on the transfer function. For simplicity, the transfer function is just analyzed in one dimension:

$$h(f_x) = \exp\left[ikd\sqrt{1-(\lambda f_x)^2}\right]. \quad (10)$$

According to the sampling theorem $|\partial h/(2\pi\partial f_x)| \le 1/2\Delta f_x$ [1], due to $|h| \le 1$, the sampling conditions of h can be written as

$$\frac{d\lambda f_x}{\sqrt{1-(\lambda f_x)^2}} \le \frac{1}{2\Delta f_x}. \quad (11)$$

In FFT, taking $\Delta f_x = 1/N\Delta x$ into Eq. (11), the upper frequency limit $f_{x_{fft\max}}$ that satisfies the sampling conditions in the FFT method can be calculated

$$f_{x_{fft\max}} = \frac{N\Delta x}{\lambda\sqrt{4d^2+N^2\Delta x^2}}. \quad (12)$$

When $S_1 = 1$ in Eq. (4), setting $f_{x_{\mathrm{MPAS}M_{\mathrm{MAX1}}}} = fx_{\mathrm{MAX\_}fft}/K_{f_x}$, the frequency sampling interval $\Delta f_x = 2f_{x_{\mathrm{MPAS}M_{\mathrm{MAX1}}}}/N$ in the matrix product method was taken into Eq. (11) to get

$$f_{x_{\mathrm{MPAS}M_{\mathrm{MAX1}}}} = \frac{\sqrt{\sqrt{(N\lambda^4)+64(N\lambda d)^2}-(N\lambda)^2}}{4\sqrt{2}d\lambda}. \quad (13)$$

Equation (13) is equivalent to the upper limit of the frequency band in [11], when $S_1 = S_2 = s\ (s \ge 1)$:

$$f_{x_{\mathrm{MPAS}M_{\mathrm{MAXS}}}} = \frac{\sqrt{\sqrt{(sN\lambda^4)+64(sN\lambda d)^2}-(sN\lambda)^2}}{4\sqrt{2}d\lambda}, \quad (14)$$

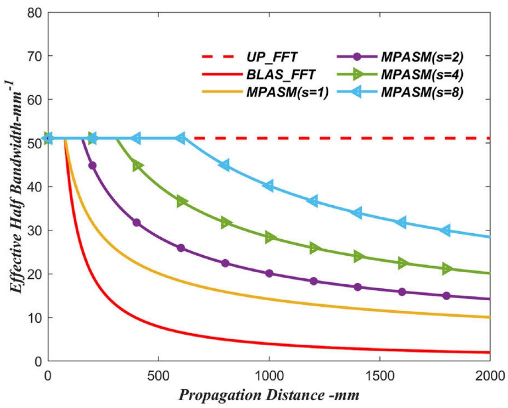

Fig. 1. Bandwidth comparison chart. The solid line is the effective bandwidth of the MPASM corresponding to different s values and BLAS. $\Delta x = 0.0098$ mm, $N = 512$, $\lambda = 632.8$ nm, and $K_{f_x} \geq 1$.

the effective bandwidth in the far field would be increased by $\sqrt{s}$ times. However, when s > 1, the computational complexity will increase, since the number of frequency domain sampling points will increase by $(s^2 - 1)MN$. The value of s needs to be selected according to the specific application. The relationship between the effective bandwidth and distance and the comparison of effective bandwidth in different methods are shown as Fig. 1.

According to Fig. 1, it is obvious that the effective bandwidth of the MPASM is much higher than that of BLAS at a long distance. This means that the calculation accuracy of the MPASM in the far field is significantly improved compared to the FFT-based BLAS. However, as the propagation distance increases, the effective bandwidth of the MPASM ($s = 1$) is also decreasing. When the distance is far enough that the effective bandwidth is lower than the bandwidth of the input wavefront, the applicable distance of the MPASM ($s = 1$) reaches its limit. At this time, the effective propagation distance can be broadened again by increasing the value of s. The following simulation will illustrate this issue in detail. Figure 2 is the simulation model, and the detailed parameters in the simulation are shown in Table 1.

When $d = 1200$ mm, the calculation results of the analytical Rayleigh–Sommerfeld (R–S) integral [1], ASM, BLAS, (CRSD) [14], and MPASM ($s = 1$) are illustrated in Fig. 3. In order to analyze the accuracy of various methods more intuitively, the simulation accuracy metric [9] is used to evaluate the accuracy of the intensity distribution:

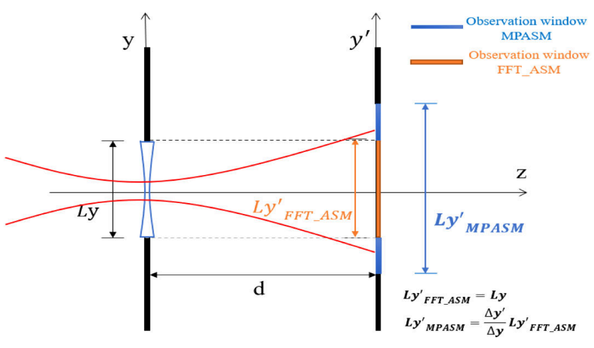

Fig. 2. Light field simulation model. Gaussian beam through lens. Ly and Ly' represent the sizes of the input and output windows, respectively.

Table 1. Parameters in Simulation

Parameter	Value
Waist radius	1 mm
Input window size	5 mm
Number of samples	512 * 512
$\Delta x, \Delta y, \Delta x', \Delta y'$	0.0098 mm
r_1, r_2	1
S_1, S_2	1
K_{f_x}, K_{f_y}	$fx_{\mathrm{MAX\_}fft}/f_{x\mathrm{MPASM}_{\mathrm{MAX1}}}$
Lens focal length	−300 mm
Wavelength	632.8 nm
CPU	core i5(2.5 GHz, RAM:16G)
GPU	NVIDIA GTX1050(2G)

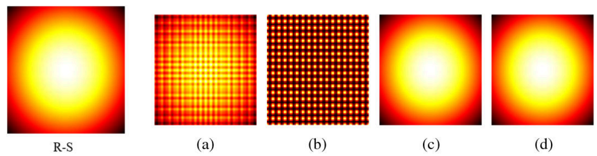

Fig. 3. Simulation results: observation plane light intensity distribution, (a) BLAS, (b) ASM, (c) CRSD, and (d) MPASM($s = 1$).

$$\mathrm{SAM} = 10lg \frac{\iint I(x, y)\,\mathrm{d}x\mathrm{d}y}{\iint |I(x, y) - \alpha I_{\mathrm{ref}}(x, y)|\,\mathrm{d}x\mathrm{d}y}, \tag{15}$$

$$\alpha = \frac{\iint I(x, y)\, I^*{}_{\mathrm{ref}}(x, y)\,\mathrm{d}x\mathrm{d}y}{\iint I_{\mathrm{ref}}(x, y)\,\mathrm{d}x\mathrm{d}y}. \tag{16}$$

To get the calculation result of the $R - S$ integral in short time, we reduce the 2-D calculation to 1-D. The reference light field I_{ref} is calculated by the $R - S$ integral.

It is well known that CRSD works well for the far field [14], and Fig. 4 verifies this. Neither CRSD nor FFT-based ASM can be applied to both near and far fields. It can be clearly seen from Fig. 4 that the MPASM maintains a fairly high calculation accuracy at both short and long distances. It is worth noting that when the propagation distance is far enough, the calculation accuracy of the MPASM ($s = 1$) will be lower than CRSD, and this trend will become more and more obvious as the distance increases. At this time, the calculation accuracy of the MPASM can be improved by increasing the value of s, as shown in Fig. 4. Through the above analysis, it can be clearly concluded that the MPASM significantly extends the effective propagation distance of the angular spectrum.

In addition, the MPASM can not only extend the effective propagation distance, but also can widen the size of the observation window without increasing the number of samples by setting the spatial sampling in Eqs. (7) and (8). As shown in Fig. 2, in FFT-based ASM, the size of the observation window is the same as the input window. However, the size of the observation window can be expanded by increasing $\Delta y'$ in Eq. (8), in the MPASM. In order to compare the difference in light intensity distribution more intuitively on the observation plane, the coma aberration in the y direction that is realized by Zernike polynomial [15] is introduced based on the model of Fig. 2. The 7th, 17th, and 29th terms in the Zernike polynomial are used in the simulation, and the corresponding coefficients are 0.8, 0.6, and -0.4.

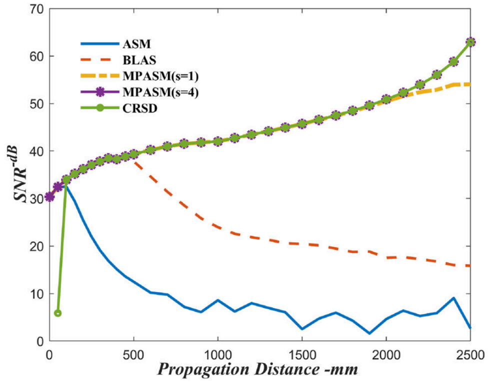

Fig. 4. Accuracy comparison chart. ASM represents the FFT-based ASM; BLAS is the FFT angle spectrum band limit method.

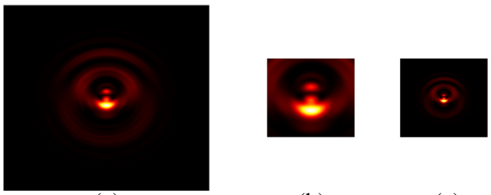

Fig. 5. Observing plane light intensity distribution, (a) $4\times$ input window CRSD (2048*2048), (b) $1\times$ input window CRSD (512*512), (c) $1\times$ input window MPASM (512*512), $\Delta y' = 4 * \Delta y$, $\Delta x' = 4 * \Delta x$.

Table 2. Calculation Time Comparison

Number of Samples	64^2	128^2	256^2	512^2	1024^2
FFT-ASM (in GPU)	0.0053 s	0.0062 s	0.0088 s	0.0103 s	0.0150 s
MPAS$M(s=1)$ (in GPU)	0.0046 s	0.0036 s	0.0069 s	0.0432 s	0.5547 s
FFT-ASM (in CPU)	0.0007 s	0.0037 s	0.0093 s	0.0335 s	0.1340 s
MPAS$M(s=1)$ (in CPU)	0.0010 s	0.0035 s	0.0148 s	0.0702 s	0.3586 s

It can be seen from Fig. 5 that the MPASM can expand the observation plane without changing the size of the input window. After analyzing the functions of the MPASM, it is also necessary to discuss calculation time-consuming issues. Matrix multiplication can be of the order $O(2N^{2.4})$ [16], and the computational complexity of the 2-D FFT is $O(N^2 \log(N^2))$. When the number of samples $N > 300$, the computational complexity of matrix multiplication is higher than that of FFT. However, due to the excellent parallel computing capabilities of GPU and powerful matrix multiplication algorithm, the MPASM based on the matrix multiplication algorithm still has a relatively fast computing speed. Table 2 shows the calculation time comparison of different sampling points in GPU and CPU [based on MATLAB], where s is equal to 1 in MPASM. The developed Python code can be found in Code 1, Ref. [17].

In conclusion, in this Letter, we propose a new method which is very concise in mathematical form for angular spectrum calculation based on the matrix product. This method uses the matrix product operation instead of FFT to avoid the problems caused by FFT sampling formulas. Based on this method, flexible sampling control that cannot be completed in FFT is implemented. The most significant advantage of the method is the flexibility of sampling control. The frequency domain sampling interval can be changed flexibly to maximize the use of all sampling points, wherefore the effective bandwidth of the far field is, and the effective calculation propagation distance of ASM has been extended. The spatial sampling of the observation window can also be controlled to alter the size of the observation window without changing the input plane. In short, the matrix product operation is used in this Letter to implement the 3-D expansion of the effective calculation scope of the ASM without changing the input plane.

Funding. National Natural Science Foundation of China (61975081, 61775097).

Disclosures. The authors declare no conflicts of interest.

REFERENCES

1. J. W. Goodman, *Introduction to Fourier Optics* (Roberts & Company, 2005).
2. O. K. Ersoy, *Diffraction, Fourier Optics and Imaging* (Wiley, 2006).
3. F. Zhang, G. Pedrini, and W. Osten, Opt. Lett. **31**, 1633 (2006).
4. C. Guo, S. Liu, and J. T. Sheridan, Appl. Opt. **54**, 4698 (2015).
5. P. Clemente, V. Durán, E. Tajahuerce, P. Andrés, V. Climent, and J. Lancis, Opt. Lett. **38**, 2524 (2013).
6. S. Pan, J. Ma, and R. Zhu, Opt. Express **25**, 20142 (2017).
7. D. G. Voelz and M. C. Roggemann, Appl. Opt. **48**, 6132 (2009).
8. J. C. Li, Z. J. Peng, and Y. C. Fu, Opt. Commun. **280**, 243 (2007).
9. X. Yu, T. Xiahui, Q. Y. Xiong, P. Hao, and W. Wei, Opt. Lett. **37**, 4943 (2012).
10. K. Matsushima and T. Shimobaba, Opt. Express **17**, 19662 (2009).
11. W. Zhang, H. Zhang, and G. Jin, Opt. Lett. **45**, 1543 (2020).
12. Y.-H. Kim, C.-W. Byun, H. Oh, J. Lee, J.-E. Pi, G. Heon Kim, M.-L. Lee, H. Ryu, H.-Y. Chu, and C.-S. Hwang, J. Opt. **16**, 125710 (2014).
13. J. D. Schmidt, *Numerical Simulation of Optical Wave Propagation with Examples in MATLAB* (SPIE, 2010).
14. I. Mamatha, T. S. B. Sudarshan, S. Tripathi, and N. Bhattar, Circuits Syst. Signal Process. **34**, 3221 (2015).
15. C. B. Duque and J. G. Sucerquia, Appl. Opt. **58**, G11 (2019).
16. S. A. Shakir, D. L. Fried, E. A. Pease, T. J. Brennan, and T. M. Dolash, Opt. Express **23**, 26853 (2015).
17. https://doi.org/10.6084/m9.figshare.12972872.